Light

GCSE Computer Science — Network Protocols (Set 1)

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A user types a website address into their browser to view a web page. Which protocol is used to request and transfer the web page?

- A. HTTP
- B. SMTP
- C. POP3
- D. FTP

Q2 (1 mark)

A user wants to send an email from their email client to a mail server. Which protocol is used for this purpose?

- ○ **A.** SMTP
- ○ **B.** HTTP
- ○ **C.** IMAP
- ○ **D.** Ethernet

Q3 (1 mark)

A website displays a padlock icon in the browser address bar.
Which protocol is being used to secure the connection?

- ○ **A.** FTP
- ○ **B.** HTTP
- ○ **C.** HTTPS
- ○ **D.** POP3

Q4 (1 mark)

A web developer needs to upload website files from their computer to a web server.
Which protocol is commonly used for this file transfer?

- ○ **A.** HTTP
- ○ **B.** SMTP
- ○ **C.** POP3
- ○ **D.** FTP

Q5 (1 mark)

An email protocol downloads messages from a mail server to a user's device and then deletes them from the server.
Which protocol is this?

- ○ **A.** POP3
- ○ **B.** SMTP
- ○ **C.** HTTP
- ○ **D.** FTP

Q6 (1 mark)

A user checks their email on both their phone and laptop. Both devices show the same emails and read status.
Which email protocol allows this synchronisation?

- ○ **A.** SMTP
- ○ **B.** FTP
- ○ **C.** IMAP
- ○ **D.** HTTP

Q7 (1 mark)

Computers in an office are connected using cables and network switches.
Which protocol governs how data is transmitted on this wired local network?

- ○ **A.** Wi-Fi

- ○ **B.** SMTP
- ○ **C.** HTTP
- ○ **D.** Ethernet

Q8 (1 mark)

A laptop connects to a network without using any cables.
Which technology enables this wireless connection?

- ○ **A.** Wi-Fi
- ○ **B.** Ethernet
- ○ **C.** FTP
- ○ **D.** POP3

Q9 (1 mark)

The TCP/IP model organises network communication into layers.
How many layers does the TCP/IP model have?

- ○ **A.** 3 layers
- ○ **B.** 4 layers
- ○ **C.** 5 layers
- ○ **D.** 7 layers

Q10 (1 mark)

A user sends an email using their email application.

At which layer of the TCP/IP model does the email application operate?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q11 (2 marks)

A router uses IP addresses to forward data packets between networks.
At which TCP/IP layer do routers primarily operate?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q12 (2 marks)

A layer of the TCP/IP model breaks data into segments and ensures reliable delivery.
Which layer is this?

- ○ **A.** Transport layer
- ○ **B.** Application layer
- ○ **C.** Internet layer

- ○ **D.** Link layer

Q13 (2 marks)

A layer of the TCP/IP model handles the physical transmission of data and uses MAC addresses.
Which layer is this?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q14 (2 marks)

A protocol suite provides the rules for how data is transmitted across the internet.
What is this protocol suite called?

- ○ **A.** HTTP/HTTPS
- ○ **B.** POP3/IMAP
- ○ **C.** TCP/IP
- ○ **D.** FTP/SFTP

Q15 (2 marks)

A school website is upgraded from HTTP to HTTPS.
What is the main benefit of this change?

- ○ **A.** The website loads faster because HTTPS is more efficient
- ○ **B.** Data transmitted between users and the website is encrypted, protecting sensitive information
- ○ **C.** The website can now display images and videos
- ○ **D.** Users no longer need passwords to access the website

Q16 (2 marks)

A user switches from POP3 to IMAP for their email.
What change will they notice?

- ○ **A.** Emails will send faster than before
- ○ **B.** Emails will be synchronised across all their devices instead of being stored on only one device
- ○ **C.** They will no longer need an internet connection to read emails
- ○ **D.** Their emails will be automatically encrypted

Q17 (2 marks)

Data travels through the TCP/IP layers when being sent across a network.
What is the correct order of layers from top to bottom?

- ○ **A.** Link, Internet, Transport, Application
- ○ **B.** Transport, Application, Link, Internet
- ○ **C.** Internet, Application, Transport, Link

- ○ **D.** Application, Transport, Internet, Link

Q18 (2 marks)

A company uses FTP to manage files on its web server.
Why might the company choose to use SFTP or FTPS instead?

- ○ **A.** To transfer files faster over the network
- ○ **B.** To encrypt file transfers and protect data from interception
- ○ **C.** To allow more users to connect simultaneously
- ○ **D.** To transfer larger files than standard FTP allows

Q19 (2 marks)

A business is choosing between Ethernet and Wi-Fi for its office network.
Which statement correctly compares these technologies?

- ○ **A.** Ethernet typically provides more stable and faster connections but requires physical cables
- ○ **B.** Wi-Fi is always faster than Ethernet for all applications
- ○ **C.** Ethernet only works for distances under 10 metres
- ○ **D.** Wi-Fi is more secure because it does not use cables that can be tapped

Q20 (2 marks)

A user's email client is configured with SMTP, POP3, and IMAP settings.
Which protocol is used when the user clicks 'Send' on a new email?

- ○ **A.** IMAP
- ○ **B.** SMTP
- ○ **C.** POP3
- ○ **D.** HTTP

Q21 (2 marks)

When data is sent across a network, each TCP/IP layer adds its own header information.
What is this process called?

- ○ **A.** Encryption
- ○ **B.** Compression
- ○ **C.** Encapsulation
- ○ **D.** Authentication

Q22 (2 marks)

At the Transport layer, TCP provides reliable delivery while UDP provides faster but unreliable delivery.
Which application would most likely use UDP?

- ○ **A.** Downloading a software update where every byte must be correct
- ○ **B.** Live video streaming where speed matters more than occasional lost data
- ○ **C.** Sending an important business email

- ○ **D.** Online banking transactions

Q23 (2 marks)

HTTP, SMTP, and FTP all operate at the same TCP/IP layer.
Which layer is this?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q24 (2 marks)

A network administrator needs to explain why protocols are necessary for network communication.
Which statement best describes why protocols are needed?

- ○ **A.** Protocols make networks run faster by reducing the amount of data sent
- ○ **B.** Protocols provide standardised rules so that different devices and software can communicate with each other
- ○ **C.** Protocols are only needed for wireless networks, not wired networks
- ○ **D.** Protocols store data on servers so it can be accessed later

Q25 (2 marks)

Different TCP/IP layers use different names for the data they handle.

Which row correctly matches layers to their data units?

- ○ **A.** Application: frames, Transport: packets, Internet: segments, Link: data
- ○ **B.** Application: data, Transport: segments, Internet: packets, Link: frames
- ○ **C.** Application: packets, Transport: frames, Internet: data, Link: segments
- ○ **D.** Application: segments, Transport: data, Internet: frames, Link: packets

Q26 (3 marks)

A user sends an email with an attachment from their computer to a colleague.
Describe the role of each TCP/IP layer in transmitting this email.

- ○ **A.** Application: adds IP addresses; Transport: creates frames; Internet: uses SMTP; Link: provides encryption
- ○ **B.** Application: uses SMTP to format the email; Transport: breaks data into segments using TCP; Internet: adds IP addresses and routes packets; Link: creates frames with MAC addresses for local delivery
- ○ **C.** Application: handles physical cables; Transport: uses Wi-Fi signals; Internet: stores the email; Link: displays the email on screen
- ○ **D.** Application: routes between networks; Transport: adds MAC addresses; Internet: formats the email; Link: uses TCP for reliability

Q27 (3 marks)

An IT manager is explaining the differences between POP3 and IMAP to staff choosing an email configuration.
Which explanation correctly compares these protocols?

○ **A.** POP3 sends emails while IMAP receives them; they perform opposite functions

○ **B.** POP3 downloads emails and removes them from the server; IMAP keeps emails on the server and synchronises across devices; both require SMTP for sending

○ **C.** POP3 is faster because it encrypts emails; IMAP is slower because it compresses them

○ **D.** POP3 works only on mobile devices; IMAP works only on desktop computers

Q28 (3 marks)

A company conducts a security audit and finds that employees are using HTTP instead of HTTPS for accessing the company intranet, and FTP instead of SFTP for file transfers.
What security risks does this create, and why?

○ **A.** No risks are created; HTTP and FTP are just as secure as their encrypted versions

○ **B.** The only risk is slower performance; unencrypted protocols use more bandwidth

○ **C.** Both HTTP and FTP transmit data in plain text, allowing attackers to intercept and read usernames, passwords, and sensitive file contents

○ **D.** The risk is that files might become corrupted; encryption prevents data corruption

Q29 (3 marks)

A network technician is diagnosing a connection problem. They determine that:

• The computer can successfully ping other devices using IP addresses
• The web browser cannot load any websites by name
• The email client cannot connect to mail servers by name
At which TCP/IP layer is the problem most likely occurring, and why?

- ○ **A.** Link layer, because the physical cables are damaged
- ○ **B.** Transport layer, because TCP connections are being blocked
- ○ **C.** Application layer, because DNS name resolution is failing and applications cannot translate domain names to IP addresses
- ○ **D.** Internet layer, because the router cannot forward packets

Q30 (3 marks)

A software company needs to choose protocols for a new application with these requirements:
• Users must log in with passwords over the internet
• Large files must be uploaded to company servers
• Real-time video calls must be supported
Which combination of protocols would best meet these requirements?

- ○ **A.** HTTP for login, FTP for uploads, TCP for video calls
- ○ **B.** HTTPS for secure login, SFTP for secure file uploads, UDP for low-latency video streaming
- ○ **C.** SMTP for login, POP3 for uploads, IMAP for video calls
- ○ **D.** Ethernet for login, Wi-Fi for uploads, HTTP for video calls

Total: **55** marks • Time: **30 min**

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